

Full Length Article

Multimodal graph-based fusion via image descriptions for few-shot open-set recognition

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ABSTRACT

Few-shot open-set recognition (FSOR) presents unique challenges due to the limited labeled samples and the presence of unknown classes during inference. While recent few-shot learning methods leverage class-level textual information to enhance feature discrimination, they often overlook image-grounded descriptions that provide more fine-grained and instance-specific semantics. Moreover, many of these approaches require prior knowledge of the class name from labeled samples during inference, which may be impractical in open-world scenarios. In this study, we propose a multimodal Graph-based Fusion (MGF) framework that learns visually grounded semantic representations to enhance FSOR performance. MGF leverages image descriptions generated by a vision-language model as the textual supervision to guide the learning of semantic features from images. A graph convolutional network is then used to fuse semantic and visual features, enabling effective intra-class information propagation and improving discrimination between known and unknown classes. We jointly optimize a contrastive and a semantic alignment loss to promote intra-class compactness and inter-class separability. Extensive experiments on several few-shot learning benchmarks demonstrate that MGF achieves superior open-set recognition and competitive closed-set classification performance.

1. Introduction

In real-world recognition tasks, collecting sufficient labeled data for every class to supervise the learning of a complete distribution is often impractical. This challenge has driven interest in few-shot learning (FSL), which focuses on predicting novel classes using a handful of labeled samples. FSL has shown promising results in domains like industrial defect detection (Du et al., 2025) and rare species identification (Dasgupta et al., 2024); however, most existing methods (Chen et al., 2023; Hou et al., 2019; Snell et al., 2017) rely on the closed-set assumption, where all test samples are assumed to belong to known classes. This assumption critically fails in open-world scenarios. For instance, a manufacturing defect classifier trained on limited fault samples may fail to reject novel defect types (e.g., material degradation patterns), causing false negatives in quality control. This highlights the need for few-shot open-set recognition (FSOR), which addresses the joint challenges of classifying known classes and rejecting unknowns.

To tackle FSOR, several methods have been proposed to integrate open-set recognition strategies into FSL paradigms. For instance, some approaches apply thresholding mechanisms (Liu et al., 2020a) or learn

compact class representations (Boudiaf et al., 2023; Sun & Chu, 2024) to separate known from unknown queries. More recent works (Huang et al., 2022; Huang & Choi, 2024b) leverage textual information for semantic augmentation. However, as shown in Fig. 1, these methods typically require textual inputs during inference on novel classes. This necessitates professional knowledge to accurately specify the class name of a novel category, which may not always be available. Moreover, incorrect class naming can introduce substantial semantic discrepancies during feature fusion, leading to feature degeneration.

In this study, we propose a novel framework called multimodal graph-based fusion (MGF) to address the challenges of FSOR by learning visually grounded semantics from image descriptions. Although vision-language models and graph neural networks have been widely explored in multimodal learning, several challenges remain insufficiently addressed in the FSOR setting. Specifically, existing VLM-based methods typically rely on semantic descriptions that may be unavailable for unseen classes, while most graph-based approaches focus primarily on improving classification accuracy with limited attention to open-set discrimination. Furthermore, the extremely scarce support samples in FSOR often lead to unstable visual-semantic alignment. Motivated by these challenges, we develop a unified framework that integrates

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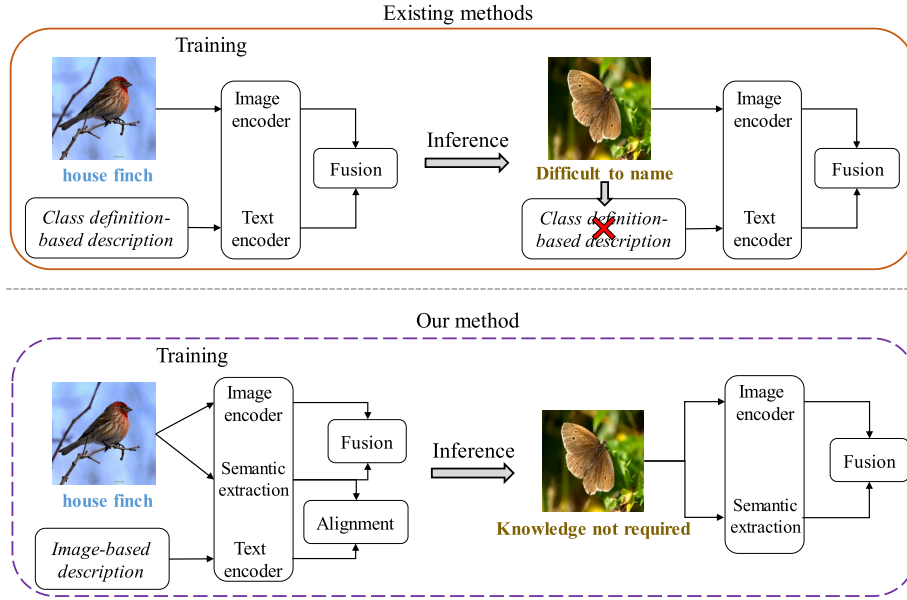


Fig. 1. Existing methods require textual inputs for novel classes during testing, which demands domain knowledge and may lead to semantic misalignment if incorrect inputs are provided. In contrast, our method employs textual supervision during training to learn semantic extraction from images, enabling text-free inference.

semantic extraction, distribution alignment, and multimodal graph fusion for few-shot open-set recognition. Instead of directly using textual information to enhance visual semantics, we leverage text as semantic supervision during training to guide the extraction of semantic features from images. This design enables our model to operate without requiring any textual input during inference (see Fig. 1). To this end, we employ a vision-language model BLIP-2 (Li et al., 2023a) to generate image-grounded descriptions using a simple yet effective prompt. These descriptions are encoded by the text encoder of CLIP (Radford et al., 2021) to produce semantic targets. For semantic feature learning, we design an encoder-decoder-based architecture with a distribution alignment loss that aligns the extracted semantic features with their textual targets while encouraging strong inter-class separability. By aligning the relational similarity distributions of image-derived semantic features and text features during training, the encoder learns semantically aligned representations from visual inputs. Consequently, the learned encoder can infer semantic-aware features directly from visual inputs during testing without requiring additional textual priors. We then perform multimodal fusion by summing the affine projections of visual and semantic features, and feed the result into a multi-layer graph convolutional network to explore and aggregate relationships among samples within each FSOR task. For effective learning, we employ a contrastive loss to help the model improve intra-class compactness while maximizing inter-class and open-set separability. We validate the effectiveness of our approach through extensive experiments on multiple few-shot benchmarks covering both FSL and FSOR tasks, along with detailed ablation studies of each proposed component. The main contributions of this study are summarized as follows:

- We propose a semantic extraction module that learns visually grounded semantic representations from image descriptions during training. This alleviates the dependence on manually provided semantic annotations for unseen classes while eliminating the need for textual input during inference.
- We introduce a multimodal graph fusion mechanism that jointly exploits visual and semantic relationships to construct task-specific sample graphs, which enhances feature aggregation and improves discrimination between known and unknown samples.
- We evaluate the proposed framework on several few-shot learning and few-shot open-set recognition benchmarks. The results show that

our method consistently achieves competitive or strong performance against existing state-of-the-art (SOTA) approaches.

2. Related work

2.1. Few-shot learning

FSL aims to learn fast adaptation to novel classes using only a few labeled samples. To this end, existing approaches explore efficient learning strategies from different perspectives, which can be roughly categorized into three types: 1) Metric-based methods (Hou et al., 2019; Kang et al., 2021; Oreshkin et al., 2018; Snell et al., 2017; Sung et al., 2018; Vinyals et al., 2016) use a metric function to learn optimal alignments between class representations and unlabeled queries in the feature space. The metric function can either be a predefined metric or parameterized as a learnable function (e.g., neural network); 2) Optimization-based methods (Chen et al., 2021; Finn et al., 2017; Lee et al., 2019; Rusu et al., 2019; Sun et al., 2019) aim to learn a good initialization or optimization rule that enables rapid adaptation to novel classes with a few samples. These methods typically achieve generalization through task-specific adaptation with minimal gradient updates; 3) Model-based methods (He et al., 2020; Liu et al., 2020b; Munkhdalai & Yu, 2017; Santoro et al., 2016) mainly incorporate architectural components (e.g., memory modules or dynamic parameters) to facilitate efficient learning from limited examples without the need for extensive fine-tuning.

2.2. Few-shot open-set recognition

FSOR extends few-shot learning by requiring the rejection of queries from unknown classes. PEELER (Liu et al., 2020a) applied entropy maximization to unknown queries to enhance feature discrimination. RFD-Net (Deng et al., 2022) leveraged pre-trained reference embeddings to alleviate randomness drift. SnaTCher (Jeong et al., 2021) measured discrepancies between transformed and modified prototype sets to reject unknown queries with large differences. SEMAN-G (Huang et al., 2022) generated negative prototypes to learn adaptive rejection boundaries for unknown queries. OPP (Sun & Chu, 2024) dynamically aggregated task-level class prototypes to classify unknown queries. While Sun and Chu (2024) and Huang et al. (2022) are the most related works to the proposed method, they focus solely on leveraging class names as textual

inputs to enhance visual features, neglecting the potential semantic gaps between visual and textual modalities. In contrast, our method utilizes image-grounded descriptions that are generated under both visual and textual guidance, providing richer and more aligned semantic supervision.

2.3. Semantic-enhanced few-shot learning

With the advent of large language models (LLMs) and vision-language models (VLMs), recent FSL methods have increasingly explored utilizing semantic augmentation to enhance feature representations. Zhang et al. (2021) leveraged attribute features to produce primitive knowledge for improving prototype representations. KSTNet (Li et al., 2023b) utilized auxiliary knowledge of class definition to enhance visual features. R-SVAE (Xu & Le, 2022) adopted text features from CLIP to synthesize more labeled samples for representation learning. BSF-PN (Huang & Choi, 2024a) dynamically assigned multimodal attention to enhance prototype representations. SP-CLIP (Chen et al., 2023) utilized class names as prompts to enhance the semantic expression of visual features, while SemFew (Zhang et al., 2024) extended the prompts by generating definitions from class names using off-the-shelf LLMs. Compared to SP-CLIP and SEM, our method generates visually grounded descriptions via induced prompts, reducing modality discrepancies. It also avoids relying on textual inputs during inference, which is beneficial in open-set or low-resource scenarios where class names are unavailable.

2.4. Graph neural network

Graph neural network (GNN) has emerged as a promising solution for reasoning about the sample-level relationships for FSL tasks, where maximizing the utilization of information propagation between limited samples is crucial. Satorras and Estrach (2018) modeled edge relations between nodes via convolutional networks and label information. EGNN (Kim et al., 2019) measured intra-class similarity and inter-class dissimilarity to learn the prediction of edge labels. TPN (Liu et al., 2019) leveraged the data manifold under transductive inference to learn a self-adaptive graph construction for label propagation. HGNN (Yu et al., 2022) proposed an instance and prototype GNN to handle inter-class distribution overlap. DPGN (Yang et al., 2020) combined sample-level and distribution-level relations to improve feature aggregation.

3. Preliminary

3.1. Few-shot learning

Few-shot learning (FSL) for image classification aims to predict labels of novel classes with only a few training samples. Let $T_i = \{S_i, Q_i^k\}_{i=1}^M$ denote a set of episodic learning tasks sampled from a base dataset D_{base} , where each task T_i consists of a support set S_i and a query set Q_i^k . Each support set $S_i = \bigcup_{n=1}^N \{(x_k^n, y_k^n)\}_{k=1}^K$ includes N classes with K labeled samples per class, where y is the corresponding label. On the other hand, each query set $Q_i^k = \{(\tilde{x}_j, \tilde{y}_j)\}_{j=1}^{Nq}$ contains the same N classes as S_i , with q unlabeled samples of each class. The value of K in the support set is typically small (e.g., one or five) to simulate the low-data regime. This setup is commonly referred to as an N -way K -shot task, where the support set provides prior knowledge for classifying the query samples. Given a learning task T_i , the goal is to learn the optimal model parameters θ^* for a model f_θ that minimizes the task-specific classification loss ℓ_{FSL} . Then, the overall training objective across all tasks can be formulated as:

$$\theta^* = \arg \min_{\theta} \sum_{i=1}^M \sum_{(\tilde{x}, \tilde{y}) \in Q_i^k} \ell_{FSL}(f_\theta(S_i, \tilde{x}), \tilde{y}). \quad (1)$$

The classification loss ℓ_{FSL} is typically instantiated as the cross-entropy loss. During inference, the learned model f_{θ^*} is evaluated on unseen tasks by leveraging support samples to classify queries.

3.2. Few-shot open-set recognition

The above definition assumes a closed-set condition, where the query and support sets share the same label space. However, in real-world scenarios, queries may belong to unknown classes not present in the support set, leading trained FSL models to misclassify unseen classes as known ones and consequently reducing overall reliability. To address this, few-shot open-set recognition (FSOR) extends the standard FSL problem by introducing an unknown query set Q_i^u into each learning task, i.e., $T_i = \{S_i, Q_i^k, Q_i^u\}$, where $Q_i^k \cap Q_i^u = \emptyset$ and Q_i^u contains samples from classes not present in the support set S_i . The goal is to correctly classify the known queries from Q_i^k while detecting and rejecting unknown ones from Q_i^u . To this end, FSOR jointly optimizes a classification loss on known queries and an open-set loss that encourages separation between known and unknown classes. The overall training objective is defined as:

$$\theta^* = \arg \min_{\theta} \sum_{i=1}^M [\ell_{FSL}^i + \lambda \cdot \ell_{FSOR}(f_\theta(S_i, Q_i^k, Q_i^u))], \quad (2)$$

where $\ell_{FSL}^i = \ell_{FSL}(f_\theta(S_i, Q_i^k))$ denotes the i th FSL loss and ℓ_{FSOR} penalizes incorrect assignment of unknown queries to known classes. The hyperparameter λ controls the trade-off between the two losses.

4. Methodology

4.1. Overview

The proposed MGF framework comprises three key modules: an image description extraction module, a semantic feature extraction module, and a graph information fusion module. Fig. 2 illustrates the overall architecture of MGF. The image description extraction module is designed to generate image descriptions that capture fine-grained visual semantics and to extract corresponding text features. Specifically, we employ the BLIP-2 model (Li et al., 2023a) with induced prompts to guide the generation of descriptions that reflect specific visual content. These descriptions are subsequently encoded using the frozen text encoder of CLIP (Radford et al., 2021), yielding text features that serve as semantic targets.

In parallel, support and query images are fed into a pre-trained backbone network to extract visual feature maps V_f . These visual features are then passed to the semantic feature extraction module, which adopts an encoder-decoder architecture to generate visually grounded semantic features $\tilde{S}_f$. To align $\tilde{S}_f$ with the text features, we introduce a distribution alignment loss that minimizes the discrepancy between their pairwise similarity distributions. The graph information fusion module takes as input the visual features $\tilde{V}_f$, which are concatenated with the corresponding semantic features $\tilde{S}_f$ to construct task-specific adjacency matrices based on pairwise feature similarities. Both visual and semantic features are linearly projected and fused to serve as node representations in the graph. A multi-layer graph convolutional network is then applied to propagate information across semantically or visually related samples. Finally, we employ a contrastive loss ℓ_{con} and a standard cross-entropy loss ℓ_{ce} to classify the known queries Q^k and reject the unknown ones Q^u .

4.2. Image description extraction

The core idea of MGF is to leverage image descriptions as a form of semantic supervision, guiding the learning of visual representations that are better aligned with class-level semantics. Existing FSL methods mainly rely on hand-crafted sentence templates as categorical cues,

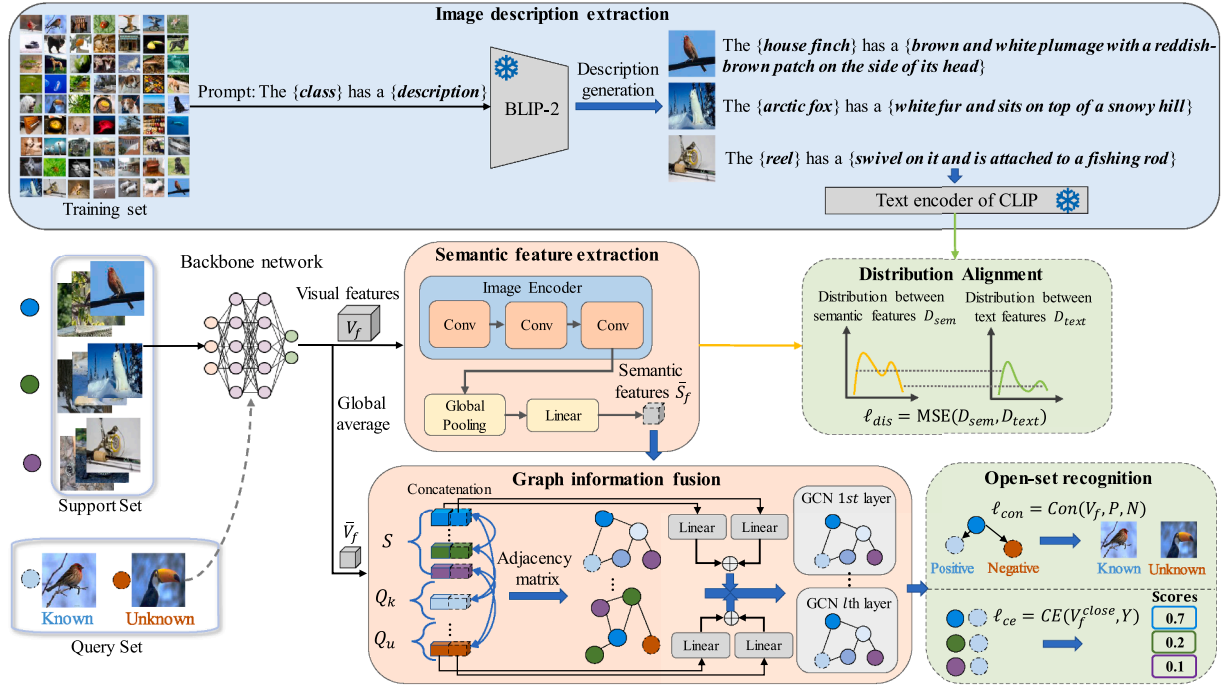


Fig. 2. The overview of the proposed multimodal graph-based fusion (MGF) framework for few-shot open-set recognition. Given a support set with class names, image descriptions are generated via a frozen vision-language model (BLIP-2) and encoded into semantic targets by a frozen text encoder of CLIP. Semantic features $\bar{S}_f$ derived from visual features V_f are aligned with semantic targets through a distribution alignment loss ℓ_{dis} . A graph-based fusion module integrates both modalities using a k -nearest neighbor-based adjacency matrix. The model is trained with the closed-set ℓ_{ce} , distribution alignment ℓ_{dis} , and open-set contrastive loss ℓ_{con} .

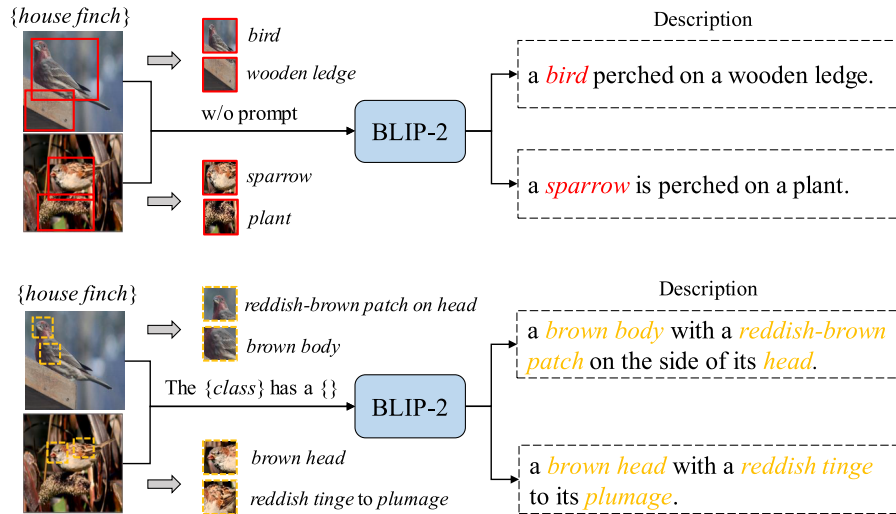


Fig. 3. Examples of text descriptions generated for the “house finch” class. Without induced prompts, BLIP-2 produces generic or inaccurate captions, which may introduce semantic ambiguity. By contrast, the proposed induced prompt encourages the model to generate class-relevant visual content, resulting in more precise and diverse descriptions.

which are then encoded using the text encoder of VLMs. For example, SP-CLIP (Chen et al., 2023) adopts the template “A photo of a {class}” to derive descriptions, while SemFew (Zhang et al., 2024) constructs prompts by combining class names with definitions generated from WordNet (Miller, 1995). However, these approaches do not consider generating text features directly from image content. Such visually grounded text features can be more naturally associated with visual features, providing semantically relevant supervision and reducing the semantic gap between modalities.

To enable text generation grounded in image content, we leverage BLIP-2 to generate visually grounded descriptions, thereby providing

semantic supervision derived directly from visual input. For instance, given images of a house finch, BLIP-2 may output descriptions like “a bird perched on a wooden ledge” or “a sparrow is perched on a plant” (see Fig. 3). However, these descriptions either provide overly generic class information (i.e., bird) or contain incorrect class names (i.e., sparrow instead of the correct house finch), resulting in ambiguity when used as semantic supervision.

We mitigate the above issue by designing an induced prompt in the format “The {class} has a {description}”, which constrains the model to generate class-relevant visual content. As illustrated in Fig. 3, the induced prompt guides BLIP-2 to focus on the visual appearance of the

object, producing descriptions such as “brown body” and “reddish-brown patch on head” for the house finch. Another advantage of using induced prompts is that when instances of the same class exhibit diverse visual traits, the generated descriptions become more varied. This diversity allows the model to better capture a broader range of visual-textual correlations, thereby enhancing the quality of semantic representation for novel classes. The generated description is concatenated with the induced prompt to form a complete sentence and is then passed through the CLIP text encoder to extract text features.

4.3. Semantic feature extraction

In FSL, the limited number of samples poses significant challenges in capturing intra-class variations and rich semantic information. Although backbone networks are capable of extracting deep visual features, the scarcity of data often hinders the model from learning class-specific semantic cues, which are essential for distinguishing between visually similar classes. This issue becomes more pronounced in open-set scenarios, where unknown classes may closely resemble known ones, leading to misclassification.

To address this issue, we introduce a semantic feature extraction module that derives high-level, image-grounded semantic representations from visual inputs. This module adopts an encoder-decoder-like structure, where the encoder first compresses visual information into a latent semantic space, and the decoder projects it into a modality-aligned feature space. This decoupled design facilitates the disentanglement of appearance and semantics and enables better alignment with language-derived concepts. It should be noted that the proposed encoder-decoder does not aim to create new semantic information beyond the visual content. Instead, it learns a mapping from visual features to semantically aligned representations under the supervision of image descriptions. Once this mapping is learned during training, the encoder can infer semantic-aware representations directly from visual inputs during inference without requiring additional textual priors. The image encoder is implemented as a stack of three convolutional layers with kernel size 3×3 , stride 1, and padding 1, each followed by a ReLU activation. This configuration maintains the spatial resolution of the input feature maps while capturing local contextual information through an expanded receptive field. The encoder receives visual feature maps $V_f \in \mathbb{R}^{C \times H \times W}$ as input and produces latent semantic features. A global average pooling layer is subsequently applied to these features, followed by a fully connected layer that projects them into the same dimension as those text features extracted by CLIP.

4.4. Graph information fusion

To enhance the discriminability of representations in FSOR, we introduce a graph-based information fusion module that integrates visual and semantic features via a graph convolutional network (GCN). FSOR tasks demand modeling relational structures among limited samples, where intra-class known samples should reinforce each other and unknown samples should be isolated from known ones. Unlike vanilla fusion methods that only perform instance-level feature aggregation, GCN enables relational reasoning through message propagation, a capability that static non-graph baselines lack and is critical for exploiting sparse sample connections.

Formally, we define a graph $\mathcal{G} = (\mathcal{X}, \mathcal{A})$, where $\mathcal{X} \in \mathbb{R}^{L \times d}$ denotes the node feature matrix for L samples, and $\mathcal{A} \in \mathbb{R}^{L \times L}$ is the adjacency matrix that reflects pairwise similarities between fused multimodal representations. Following ProtoNet, we compute class prototypes for visual and semantic features by averaging the support features from the same class, which can reduce the complexity of the graph structure and improve robustness. To construct the adjacency matrix, the visual features $\tilde{V}_f$ and semantic features $\tilde{S}_f$ are first concatenated:

$$\tilde{\mathbf{X}} = [\tilde{V}_f \parallel \tilde{S}_f] \in \mathbb{R}^{L \times 2d}. \quad (3)$$

Concatenation preserves the full joint information of both modalities, which is essential for determining whether two samples should be connected. We then compute pairwise similarities between all sample pairs:

$$S_{ij} = \frac{\tilde{\mathbf{X}}_i^\top \tilde{\mathbf{X}}_j}{\|\tilde{\mathbf{X}}_i\|_2 \cdot \|\tilde{\mathbf{X}}_j\|_2}, \quad \forall i, j \in \{1, \dots, L\}. \quad (4)$$

For each sample, we select its top- k nearest neighbors to form the edges in the adjacency matrix. To reduce the risk of incorrect aggregation caused by unknown samples being overly close to known samples, we explicitly include a self-loop for each node in the graph. That is, each node is connected to its k nearest neighbors and itself:

$$\mathcal{N}_k(i) = \{j \mid j \in \text{TopK}(S_{ij}, k + 1)\}, \quad (5)$$

where $\text{TopK}(\cdot)$ returns the indices corresponding to the top $k + 1$ highest similarity values, and $\mathcal{N}_k(i)$ denotes the neighbor set of node i . Through self-looping, the model preserves the information of each node, preventing drastic changes during the message propagation process. As a result, each row in the adjacency matrix $\mathcal{A}$ contains $k + 1$ non-zero entries, corresponding to the k neighbors and the self-loop. The adjacency matrix $\mathcal{A}$ can be formalized as:

$$A_{ij} = \begin{cases} 1, & \text{if } j \in \mathcal{N}_k(i) \\ 0, & \text{otherwise,} \end{cases} \quad (6)$$

$$\tilde{A}_{ij} = \frac{A_{ij}}{\sum_j A_{ij} + \epsilon}. \quad (7)$$

$\tilde{A}_{ij}$ denotes the row-normalized adjacency matrix used for message propagation, and ϵ is a small constant added for numerical stability to avoid division by zero.

For node features $\mathcal{X}$ used in the graph convolutional network, we first map $\tilde{V}_f$ and $\tilde{S}_f$ into a shared embedding space via separate linear layer, denoted as $\tilde{V}'_f$ and $\tilde{S}'_f$, and then aggregate them via element-wise addition:

$$\mathcal{X} = \tilde{V}'_f + \tilde{S}'_f \in \mathbb{R}^{L \times d}. \quad (8)$$

This design choice is motivated by the need for efficient and compact feature fusion. Addition preserves the original dimensionality, avoiding the curse of dimensionality in graph convolution layers. This is especially critical in few-shot scenarios where over-parameterization easily leads to overfitting. Once the projection layers align the two modalities, addition naturally aggregates complementary information. In contrast, concatenation would double the input dimension, increasing parameters and computational cost without necessarily improving fusion performance.

Then, the GCN is applied to propagate and refine the node features over the constructed graph. Following the standard formulation, the graph convolution operation is defined as follows:

$$\mathcal{X}^{(l+1)} = \sigma(\tilde{A} \mathcal{X}^{(l)} \mathbf{W}^{(l)}). \quad (9)$$

$\mathcal{X}^{(l+1)}$ denotes the node representation at the l th layer, $\mathbf{W}^{(l)}$ is the learnable weight matrix, and σ is a non-linear activation function. The initial input is set to $\mathcal{X}^{(0)} = \mathcal{X}$. After L_g layers of propagation, we obtain the final node embeddings $\mathcal{Z} = \mathcal{X}^{(L_g)}$, which are used for the subsequent open-set recognition tasks.

4.5. Loss optimization

FSOR tasks require careful loss design to balance the classification of known classes and rejection of unknown ones. To this end, we first apply a standard cross-entropy loss ℓ_{ce} to supervise the classification of known query features $\mathcal{Z}_{Q^k}$. For each known query feature $q_{n,i}^k \in \mathcal{Z}_{Q^k}$, the model predicts its class label $\hat{y}_i^k$ based on the prototype embedding $P_j \in \mathcal{Z}_S$, where $\mathcal{Z}_S \in \mathbb{R}^{N \times d}$. The loss is defined as:

$$\ell_{ce} = - \sum_{i \in \mathcal{Z}_{Q^k}} \sum_{n=1}^N 1(\hat{y}_i^k = n) \log \hat{p}_{(n)}^k, \quad (10)$$

$$\tilde{p}_i^k(n) = \frac{\exp\left(s(q_{n,i}^k, P_n)\right)}{\sum_{j=1}^N \exp\left(s(q_{n,i}^k, P_j)\right)}. \quad (11)$$

$\tilde{p}_i^k$ indicates the predicted probability for class n , $s(\cdot)$ denotes the cosine similarity, and $\mathbb{1}(\cdot)$ is the indicator function.

In FSOR tasks, it is crucial that prototype embeddings exhibit high similarity to queries from known classes while maintaining low similarity to queries from unknown classes. To encourage this behavior during training, we adopt an InfoNCE-based contrastive loss (Shi et al., 2022), which simultaneously reinforces prototype alignment with known queries and penalizes their similarity to unknown queries. Specifically, for each known query $q_{n,i}^k$ from class n , we define a positive pair with the corresponding prototype P_n . To construct a meaningful contrast, the negative terms are formed by computing the similarities between the same prototype P_n and all unknown queries $q_j^u \in \mathcal{Z}_{Q^u}$. The contrastive loss ℓ_{con} is defined as:

$$\ell_{\text{con}} = -\frac{1}{|\mathcal{Z}_{Q^k}|} \sum_{n=1}^N \sum_{i=1}^q \log \frac{\exp\left(s(P_n, q_{n,i}^k)/\tau\right)}{D_{n,i}}, \quad (12)$$

where the denominator term $D_{n,i}$ is given by:

$$D_{n,i} = \exp\left(s(P_n, q_{n,i}^k)/\tau\right) + \sum_{q_j^u \in \mathcal{Z}_{Q^u}} \exp\left(s(P_n, q_j^u)/\tau\right). \quad (13)$$

Here, τ denotes a temperature hyperparameter that controls the sharpness of the similarity distribution. Unlike standard InfoNCE loss that balances the number of positive and negative samples, our formulation reflects the asymmetry in FSOR tasks: each prototype is paired only with the known queries belonging to the same class (resulting in fewer positive pairs), but contrasted against all unknown queries, regardless of class. To account for this imbalance, our formulation aggregates all unknown queries into a single negative term by summing their similarities with the prototypes. This design choice treats the negative contribution as a global contrastive signal, encouraging the positive pairs to suppress all the similarities between the unknown queries and prototypes.

For bridging the gap between semantic and text features, we formulate a distribution alignment loss ℓ_{dis} , which calculates pairwise similarities within each modality and minimizes the alignment difference between modalities. Specifically, let $\tilde{S}_f \in \mathbb{R}^{L \times d}$ denote the matrix of semantic features extracted from the vision branch, and $T_{\text{txt}} \in \mathbb{R}^{L \times d}$ denote the matrix of text features obtained from the image description extraction process. For each sample i ($1 \leq i \leq L$), $\tilde{S}_f[i] \in \mathbb{R}^d$ represents the semantic feature vector of the i th sample, and $T_{\text{txt}}[i] \in \mathbb{R}^d$ represents the corresponding text feature vector. We construct a similarity distribution for each feature by measuring its relation to all other features within the same modality, and the loss is computed as:

$$\ell_{\text{dis}} = \frac{1}{L} \sum_{i=1}^L \left\| \text{sim}(\tilde{S}_f[i], \tilde{S}_f) - \text{sim}(T_{\text{txt}}[i], T_{\text{txt}}) \right\|_2^2. \quad (14)$$

Here, $\text{sim}(\tilde{S}_f[i], \tilde{S}_f)$ denotes the similarity vector between the i th semantic feature and all semantic features in $\tilde{S}_f$, defined as

$$\text{sim}(\tilde{S}_f[i], \tilde{S}_f) = [\cos(\tilde{S}_f[i], \tilde{S}_f[j])]_{j=1}^L,$$

where $\cos(\cdot, \cdot)$ denotes the cosine similarity between two vectors. The same definition applies to $\text{sim}(T_{\text{txt}}[i], T_{\text{txt}})$.

By minimizing the discrepancy between these distributions across modalities, we encourage alignment between semantic and textual representations. The final loss function combines the closed-set classification loss ℓ_{ce} , the prototype-based contrastive loss ℓ_{con} , and the distribution alignment loss ℓ_{dis} to jointly optimize the model. The overall objective is given by:

$$\mathcal{L} = \ell_{\text{ce}} + \lambda \ell_{\text{con}} + \ell_{\text{dis}}, \quad (15)$$

where λ is a balancing hyperparameter. The closed-set classification loss and the alignment loss have comparable magnitudes during optimization; therefore, we do not apply additional weighting between them.

Through joint optimization, the model learns to correctly construct the adjacency matrix and obtain more discriminative features through information fusion between modalities, thereby accurately classifying known classes and rejecting unknown classes.

Algorithm 1 Overview of the proposed MGF framework.

Require: Episode images $\mathcal{X}$, text descriptions $\mathcal{T}$, visual encoder f_θ , semantic projector g_ϕ , graph aggregator h_ψ

- 1: **Forward pass:**
 - 2: $\mathbf{F} \leftarrow f_\theta(\mathcal{X})$
 - 3: $\tilde{V}_f \leftarrow \text{GAP}(\mathbf{F})$, $\tilde{S}_f \leftarrow g_\phi(\mathbf{F})$ ▷ semantic extraction
 - 4: $\tilde{\mathbf{X}} \leftarrow [\tilde{V}_f \| \tilde{S}_f]$ ▷ multimodal representation
 - 5: $\tilde{\mathcal{A}} \leftarrow \text{TopK}(\cos(\tilde{\mathbf{X}}))$ ▷ graph construction
 - 6: $\mathcal{Z} \leftarrow h_\psi(\tilde{V}_f, \tilde{S}_f, \tilde{\mathcal{A}})$ ▷ graph information fusion
 - 7: **Backward pass:**
 - 8: $\mathcal{L} \leftarrow \ell_{\text{ce}} + \lambda \ell_{\text{con}} + \ell_{\text{dis}}$ ▷ joint optimization
 - 9: Update (θ, ϕ, ψ)
 - 10: **Inference:**
 - 11: $\mathcal{Z} \leftarrow h_\psi(\tilde{V}_f, g_\phi(f_\theta(\mathcal{X})), \tilde{\mathcal{A}})$
 - 12: $\hat{\mathcal{Y}} \leftarrow \text{OpenSetClassification}(\cos(\mathcal{Z}_Q, \text{Proto}(\mathcal{Z}_s)))$
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Algorithm 1 summarizes the overall pipeline of the proposed MGF framework. In each episode, visual and text features are extracted and fused through graph propagation to produce discriminative embeddings. The model is trained to jointly minimize classification error, separate known and unknown samples, and align visual-text distributions, following the objective in Eq. (15). During inference, the model directly infers semantic features from images without external text descriptions, which is particularly beneficial for open-set scenarios where text priors may be unavailable.

5. Experiments

5.1. Datasets

This study evaluates the proposed method on five few-shot learning benchmarks, including mini-ImageNet (Vinyals et al., 2016), tiered-ImageNet (Ren et al., 2018), CUB-200-2011 (Wah et al., 2011), FC100 (Oreshkin et al., 2018), and CIFAR-FS (Bertinetto et al., 2018). Mini-ImageNet is a subset of ImageNet (Deng et al., 2009) with 100 classes and 600 images per class, commonly split into 64/16/20 classes for training, validation, and testing. Tiered-ImageNet is another variant that consists of 608 ImageNet classes, with 351 training, 97 validation, and 160 testing classes. CUB-200-2011 is a fine-grained dataset that includes 200 bird classes, using 100/50/50 classes for training, validation, and testing. CIFAR-FS follows the same criteria as mini-ImageNet to split the dataset into 64/16/20 classes, while FC100 adopts a 64/20/20 split with additional constraints based on superclass hierarchy to increase task difficulty.

Following previous work (Huang et al., 2022; Jeong et al., 2021; Wang et al., 2023), we adopt the classification accuracy (ACC) and Area Under the Receiver Operating Characteristics (AUROC) to assess the performance of FSOR. ACC measures the performance of closed-set classification, whereas AUROC evaluates the ability to distinguish between known and unknown classes across varying thresholds. In addition, AUPRC, FPR95, and F1-score are also adopted to assess the effectiveness of each proposed component. AUPRC measures the area under the precision-recall curve, FPR95 reflects the proportion of false positive rate when the true positive rate reaches 95%, and F1-score is defined as the harmonic mean of precision and recall.

5.2. Implementation details

Experiments were conducted using PyTorch 1.12 on an RTX 3090 Ti GPU. We adopted ResNet-12 as the backbone and pre-trained it on

Table 1

Few-shot open-set recognition results (%) on mini-ImageNet and tiered-ImageNet. ACC and AUROC denote model performances on the closed- and open-set tasks, respectively.

Method	mini-ImageNet 5-way				tiered-ImageNet 5-way			
	1-shot		5-shot		1-shot		5-shot	
	ACC	AUROC	ACC	AUROC	ACC	AUROC	ACC	AUROC
OpenMax (Bendale & Boult, 2016)	63.69 ± 0.84	62.64 ± 0.80	80.56 ± 0.58	62.27 ± 0.71	68.28 ± 0.95	60.13 ± 0.74	83.48 ± 0.66	65.51 ± 0.83
ProtoNet (Snell et al., 2017)	64.01 ± 0.88	51.81 ± 0.93	80.09 ± 0.58	60.39 ± 0.92	68.26 ± 0.96	60.73 ± 0.80	83.40 ± 0.65	64.96 ± 0.83
CounterFactual (Neal et al., 2018)	63.70 ± 0.83	64.17 ± 0.88	81.44 ± 0.54	71.58 ± 0.76	70.08 ± 0.94	71.04 ± 0.80	85.36 ± 0.60	78.66 ± 0.62
FEAT (Ye et al., 2020)	67.02 ± 0.85	57.01 ± 0.84	82.02 ± 0.53	63.18 ± 0.78	70.52 ± 0.96	63.54 ± 0.76	84.74 ± 0.64	70.74 ± 0.75
PEELER (Liu et al., 2020a)	65.86 ± 0.85	60.57 ± 0.83	80.61 ± 0.59	67.35 ± 0.80	69.51 ± 0.92	65.20 ± 0.76	84.10 ± 0.66	73.27 ± 0.71
RFDNet (Deng et al., 2022)	66.40 ± 0.82	71.91 ± 0.78	81.19 ± 0.60	81.29 ± 0.56	70.33 ± 0.90	75.12 ± 0.74	84.88 ± 0.62	82.19 ± 0.66
SnaTCHer-L (Jeong et al., 2021)	67.60 ± 0.83	69.40 ± 0.92	82.36 ± 0.58	76.15 ± 0.83	70.85 ± 0.99	74.95 ± 0.83	85.23 ± 0.64	80.81 ± 0.68
REFOCS (Nag et al., 2023)	66.35 ± 0.87	70.68 ± 0.60	82.44 ± 0.98	76.83 ± 0.60	70.61 ± 0.92	75.12 ± 0.76	85.34 ± 0.57	81.16 ± 0.72
SEMAN-G (Huang et al., 2022)	68.24 ± 0.82	72.85 ± 0.69	83.48 ± 0.48	82.07 ± 0.58	71.06 ± 0.92	74.27 ± 0.77	86.02 ± 0.58	82.59 ± 0.57
ASOP-L (Kim et al., 2023)	67.85 ± 0.20	71.91 ± 0.70	82.81 ± 0.30	81.04 ± 0.20	71.49 ± 0.40	75.04 ± 0.40	85.15 ± 0.10	81.51 ± 0.10
OSLO (Boudiaf et al., 2023)	71.73 ± 0.72	74.92 ± 0.79	83.40 ± 0.44	82.59 ± 0.73	76.79 ± 0.74	79.06 ± 0.78	85.28 ± 0.52	80.43 ± 0.68
MRM _r -SnaTCHerF (Che et al., 2023)	67.03 ± 0.83	71.20 ± 0.80	82.00 ± 0.55	80.39 ± 0.59	71.13 ± 0.91	75.59 ± 0.77	85.27 ± 0.62	83.03 ± 0.63
OPP ^a (Sun & Chu, 2024)	68.52 ± 0.70	72.68 ± 0.68	83.57 ± 0.55	81.76 ± 0.60	73.06 ± 0.91	75.61 ± 0.77	86.68 ± 0.58	83.31 ± 0.72
GEL (Wang et al., 2023)	68.26 ± 0.85	73.70 ± 0.82	83.05 ± 0.55	82.29 ± 0.60	70.50 ± 0.93	75.86 ± 0.81	84.60 ± 0.65	81.95 ± 0.72
BiDirP (Zhang et al., 2025)	68.62 ± 0.85	74.94 ± 0.86	83.12 ± 0.52	83.94 ± 0.61	71.36 ± 0.94	76.28 ± 0.78	85.28 ± 0.68	84.43 ± 0.63
TNPNet (Wu et al., 2025)	68.22 ± 0.87	74.21 ± 0.79	82.50 ± 0.82	84.07 ± 0.82	71.61 ± 0.90	78.11 ± 0.77	84.96 ± 0.63	86.56 ± 0.53
ASCM ^a (Li et al., 2026)	68.27 ± 0.69	76.16 ± 0.68	83.25 ± 0.55	84.32 ± 0.60	71.35 ± 0.76	77.18 ± 0.70	86.33 ± 0.55	83.62 ± 0.45
MGF (Ours)	74.59 ± 0.77	78.02 ± 0.70	82.78 ± 0.50	87.35 ± 0.40	77.26 ± 0.78	81.86 ± 0.69	85.29 ± 0.54	88.67 ± 0.45

^a denotes our implementations. Results are reported with 95% confidence intervals and the best results are highlighted in bold.

the base dataset following Huang and Choi (2024b). Tiered-ImageNet was trained for 90 and 300 epochs for the remaining datasets using the SGD optimizer with a learning rate of 0.1. Cosine annealing with warm restarts was applied to adjust the learning rate. We used a 2-layer GNN with ReLU activations for feature fusion. The hyperparameters λ and τ were selected based on validation experiments and set to 0.8 and 0.5, respectively. Image descriptions were generated using BLIP-2 with Flan-T5-XL as the language model (Chung et al., 2024). The maximum generation length was set to 70 tokens to satisfy the input length constraint of the CLIP text encoder. The generated descriptions were subsequently encoded by the frozen CLIP text encoder to obtain text features. In the meta-training, the proposed method was trained for 30 epochs with 400 episodes, with learning rates of 1e-6 for the backbone and 5e-4 for the proposed modules. Following the general setup in previous work (Sun & Chu, 2024; Wang et al., 2023), each class was assigned 15 known queries, and the unknown query set consisted of 5 negative classes with 15 unknown queries per class. We randomly sampled 1000 episodes to evaluate the performance on 5-way 1-shot and 5-shot FSOR tasks, and the averaged results are reported with 95% confidence intervals.

5.3. Main results

Results on mini- and tiered-ImageNet: Table 1 presents the comparison results on mini-ImageNet and tiered-ImageNet. On the mini-ImageNet benchmark, MGF achieves the highest AUROC scores of 78.02% in the 1-shot and 87.35% in the 5-shot settings for open-set recognition, outperforming the semantic-based method SEMAN-G by 5.17% and 5.28%, respectively. Compared with recent FSOR methods such as BiDirP, TNPNet, and ASCM, MGF further improves AUROC scores by 1.86% and 3.03% under the 1-shot and 5-shot settings, respectively. OPP achieves the best closed-set accuracy of 83.57% in the 5-shot setting by learning a task-specific positive prototype, while MGF achieves the highest 1-shot accuracy of 74.59%. On tiered-ImageNet, MGF achieves the highest AUROC score of 81.86% in the 1-shot and 88.67% in the 5-shot settings, significantly outperforming competing approaches. Moreover, MGF surpasses the recently proposed TNPNet by 3.75% and 2.11% in AUROC under the 1-shot and 5-shot settings, respectively. MGF demonstrates strong competitiveness in closed-set recognition, achieving 77.26% in the 1-shot setting and outperforming the transductive method OSLO. These results demonstrate that MGF

maintains a strong capability in distinguishing known and unknown classes.

Results on CIFAR-FS and FC100: Table 2 presents the comparison results on CIFAR-FS and FC100. On the CIFAR-FS benchmark, MGF achieves the highest AUROC scores of 83.36% in the 1-shot setting and 89.34% in the 5-shot setting, along with the best closed-set accuracies of 81.35% and 87.37%, respectively. Compared with recent FSOR methods such as BiDirP, TNPNet, and ASCM, MGF further improves the best competing AUROC by 3.20% and 1.55% under the 1-shot and 5-shot settings, respectively. In particular, compared to the strong semantic-based baseline SEMAN-G, MGF brings a notable improvement of 5.26% in AUROC and 6.80% in ACC under the 1-shot setting. On FC100, MGF also achieves the best AUROC performance among all competing methods. It achieves the highest AUROC scores of 61.55% in the 1-shot and 67.76% in the 5-shot settings, while also attaining competitive closed-set performance. Compared with the recent ASCM method, MGF further improves AUROC by 0.88% and 1.44% under the 1-shot and 5-shot settings, respectively. Despite the low image resolution (i.e., 32×32), MGF exhibits strong generalizability compared to SOTA methods such as OPP, GEL, BiDirP, TNPNet, and ASCM by effectively fusing multimodal information to enhance feature semantics.

Results on CUB-200-2011: As shown in Table 3, our method achieves the best results across both 1-shot and 5-shot settings, significantly outperforming all previous approaches. Specifically, MGF obtains an AUROC of 92.10% and accuracy of 89.35% in the 1-shot setting, and further improves to 96.45% AUROC and 93.47% accuracy in the 5-shot setting. Compared with SEMAN-G, MGF achieves gains of 10.06% in AUROC and 7.69% in accuracy under 1-shot, and 7.11% in AUROC and 2.45% in accuracy under 5-shot. These results indicate the strong generalizability of MGF in handling subtle inter-class differences and high intra-class similarities in fine-grained tasks.

Discussion: Although MGF achieves slightly lower ACC than some baselines in a few settings (e.g., miniImageNet 5-shot and FC100 1-shot), this behavior is closely related to the objective of FSOR. Unlike conventional few-shot classification methods that focus solely on closed-set recognition, MGF jointly optimizes classification, distribution alignment, and open-set contrastive objectives. In particular, the open-set contrastive loss explicitly pushes known prototypes away from unknown query embeddings to improve category separability. While this additional constraint enhances the detection of unknown samples, it may

Table 2

Few-shot open-set recognition results (%) on CIFAR-FS and FC100. ACC and AUROC denote model performances on the closed- and open-set problems, respectively.

Method	CIFAR-FS 5-way				FC100 5-way			
	1-shot		5-shot		1-shot		5-shot	
	ACC	AUROC	ACC	AUROC	ACC	AUROC	ACC	AUROC
OpenMax (Bendale & Boulton, 2016)	71.65 ± 0.65	50.21 ± 0.70	85.66 ± 0.48	75.78 ± 0.47	44.70 ± 0.73	50.10 ± 0.65	60.11 ± 0.66	57.78 ± 0.56
ProtoNet ^a (Snell et al., 2017)	64.01 ± 0.88	51.81 ± 0.93	80.09 ± 0.58	60.39 ± 0.92	40.10 ± 0.74	57.36 ± 0.70	55.12 ± 0.72	62.05 ± 0.66
CounterFactual (Neal et al., 2018)	71.71 ± 0.65	72.57 ± 0.61	85.71 ± 0.45	80.44 ± 0.37	44.53 ± 0.70	57.20 ± 0.65	61.12 ± 0.66	62.35 ± 0.56
PEELER (Liu et al., 2020a)	71.47 ± 0.67	71.28 ± 0.57	85.46 ± 0.47	75.97 ± 0.33	44.45 ± 0.70	55.86 ± 0.65	60.86 ± 0.66	61.07 ± 0.56
RFDNet (Deng et al., 2022)	73.83 ± 0.92	73.35 ± 0.77	85.12 ± 0.74	84.40 ± 0.64	41.00 ± 0.75	57.90 ± 0.58	57.38 ± 0.74	62.49 ± 0.56
SnaTCHer-F (Jeong et al., 2021)	75.09 ± 0.87	78.15 ± 0.77	87.18 ± 0.62	85.81 ± 0.64	44.68 ± 0.69	59.54 ± 0.67	60.24 ± 0.69	63.16 ± 0.61
SEMAN-G (Huang et al., 2022)	74.55 ± 0.85	78.10 ± 0.79	86.71 ± 0.67	86.47 ± 0.67	46.01 ± 0.70	59.73 ± 0.65	62.18 ± 0.66	64.46 ± 0.56
OPP ^a (Sun & Chu, 2024)	75.16 ± 0.65	78.37 ± 0.77	87.21 ± 0.55	85.36 ± 0.71	41.75 ± 0.76	57.51 ± 0.65	61.58 ± 0.74	65.39 ± 0.65
GEL ^a (Wang et al., 2023)	76.77 ± 0.88	78.67 ± 0.80	86.74 ± 0.66	86.56 ± 0.59	44.20 ± 0.76	58.49 ± 0.65	60.37 ± 0.74	62.24 ± 0.64
BiDirP ^a (Zhang et al., 2025)	75.32 ± 0.77	79.18 ± 0.66	86.87 ± 0.51	87.79 ± 0.46	44.35 ± 0.71	59.86 ± 0.62	59.92 ± 0.63	65.68 ± 0.56
TNPNet ^a (Wu et al., 2025)	73.94 ± 0.83	79.35 ± 0.73	85.18 ± 0.58	87.65 ± 0.54	43.61 ± 0.72	59.37 ± 0.62	58.78 ± 0.63	65.81 ± 0.55
ASCM ^a (Li et al., 2026)	76.85 ± 0.75	80.16 ± 0.70	83.25 ± 0.55	87.62 ± 0.45	45.32 ± 0.70	60.67 ± 0.60	61.03 ± 0.65	66.32 ± 0.58
MGF (Ours)	81.35 ± 0.76	83.36 ± 0.66	87.37 ± 0.50	89.34 ± 0.45	45.75 ± 0.72	61.55 ± 0.61	61.80 ± 0.63	67.76 ± 0.56

^a denotes our implementations. Results are reported with 95% confidence intervals and the best results are highlighted in bold.

Table 3

Few-shot open-set recognition results (%) on CUB-200-2011. ACC and AUROC denote model performances on the closed- and open-set problems, respectively. All comparative methods are re-implemented under the same experimental setup. The results are reported with 95% confidence intervals and the best results are highlighted in bold.

Method	CUB-200-2011 5-way			
	1-shot		5-shot	
	ACC	AUROC	ACC	AUROC
ProtoNet (Snell et al., 2017)	71.55 ± 0.88	74.67 ± 0.71	86.91 ± 0.51	80.35 ± 0.52
FEAT (Ye et al., 2020)	75.43 ± 0.77	76.56 ± 0.70	87.49 ± 0.52	86.3 ± 0.52
FRN (Wertheimer et al., 2021)	80.51 ± 0.88	77.19 ± 0.65	89.78 ± 0.47	85.02 ± 0.49
PEELER (Liu et al., 2020a)	72.33 ± 0.76	73.46 ± 0.63	86.28 ± 0.47	83.55 ± 0.44
RFDNet (Deng et al., 2022)	75.37 ± 0.85	78.92 ± 0.70	88.97 ± 0.53	87.47 ± 0.53
MRM _T -SnaTCHerF (Che et al., 2023)	70.16 ± 0.85	76.28 ± 0.70	84.67 ± 0.52	85.76 ± 0.50
SnaTCHer-F (Jeong et al., 2021)	73.54 ± 0.88	76.64 ± 0.79	86.60 ± 0.59	83.89 ± 0.57
OPP (Sun & Chu, 2024)	77.49 ± 0.77	79.68 ± 0.70	89.37 ± 0.52	88.51 ± 0.52
SEMAN-G (Huang et al., 2022)	81.66 ± 0.83	82.04 ± 0.74	91.02 ± 0.47	89.34 ± 0.49
GEL (Wang et al., 2023)	76.01 ± 0.91	79.33 ± 0.72	89.14 ± 0.52	87.74 ± 0.51
MGF (Ours)	89.35 ± 0.59	92.10 ± 0.45	93.47 ± 0.29	96.45 ± 0.20

slightly reduce the compactness of known-class representations and alter decision boundaries among known classes, leading to a minor reduction in ACC. For example, on miniImageNet 5-shot, MGF achieves 82.78% ACC compared with 83.57% for OPP, while improving AUROC from 81.76% to 87.35%.

On FC100, the limited image resolution (32×32) reduces the quality of BLIP-2-generated descriptions and weakens the semantic supervision available for the semantic extraction module. Consequently, the benefit of semantic guidance to fine-grained closed-set classification becomes less pronounced. Nevertheless, the proposed cross-modal alignment and graph-based constraints remain effective in separating known and unknown categories, resulting in consistently strong open-set discrimination. Similar trends can also be observed in other settings, where marginal reductions in ACC are accompanied by notable gains in AUROC, highlighting the effectiveness of MGF in balancing closed-set recognition and unknown-category detection.

5.4. Ablation studies

5.4.1. Analysis of core modules

Table 4 reports the ablation results on mini-ImageNet and tiered-ImageNet under both the 5-way 1-shot and 5-way 5-shot settings. Compared with the baseline, introducing the graph information fusion (GIF) module consistently improves all evaluation metrics. By exploiting top- k nearest-neighbor relationships, GIF increases AUROC from 68.68% to 76.60% on mini-ImageNet and from 75.96% to 79.95% on tiered-ImageNet in the 1-shot setting, while reducing FPR95 by 9.75% and 11.69%, respectively. Similar improvements are observed in the 5-shot setting, where AUROC increases by 6.33% and 5.37%. Further incorporating the semantic extraction (SE) module leads to additional gains across both datasets and shot settings. In particular, AUROC improves from 76.60% to 78.02% on mini-ImageNet and from 79.95% to 81.86% on tiered-ImageNet in the 1-shot setting, and from 86.06% to 87.35% and 87.73% to 88.67% in the 5-shot setting. Similar trends can be observed for FPR95, AUPRC, and F1-Score. These results suggest that graph-based feature aggregation enhances the modeling of sample relationships, while semantic guidance further improves graph construction and feature discrimination.

5.4.2. Analysis of semantic feature effectiveness

To investigate whether the performance improvement originates from the learned semantic features or merely from the increased feature dimensionality, we conduct two additional control experiments. In the first setting (Random), the semantic extraction module is replaced with a fixed random projection. In the second setting (VisualOnly), it is replaced with a learnable fully connected layer. Both settings preserve the same 512-dimensional feature space, allowing the contribution of the proposed semantic extraction module to be evaluated independently of feature dimensionality.

As shown in Table 5, both variant configurations consistently perform worse than the proposed method across all datasets and evaluation settings. For instance, on the mini-ImageNet 5-shot task, the Random and VisualOnly variants achieve accuracies of 80.92% and 81.98%, respectively, whereas the proposed method attains 82.78%. A similar pattern is observed in AUROC scores, with values of 84.06%, 85.76%, and 87.35% for the Random, VisualOnly, and proposed approaches, respectively. These improvements are also consistent on tiered-ImageNet and CUB. Notably, VisualOnly outperforms Random across all cases, suggesting that learning an additional visual projection is more beneficial than relying on a random high-dimensional representation. Nevertheless, the proposed method consistently outperforms VisualOnly, indicating that

Table 4

Few-shot open-set recognition results (%) of the core module analysis on mini-ImageNet and tiered-ImageNet under the 5-way 1-shot and 5-shot settings.

Dataset	Shot	Setting	ACC $\uparrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUPRC $\uparrow$	F1 $\uparrow$
mini-ImageNet	1-shot	Baseline ^a	68.68	68.68	74.01	71.68	66.07
		+ GIF	72.52 (+3.84)	76.60 (+7.92)	64.26 (-9.75)	76.67 (+4.99)	70.14 (+4.07)
		+ SE	74.59 (+5.91)	78.02 (+9.34)	62.04 (-11.97)	78.14 (+6.46)	70.86 (+4.79)
	5-shot	Baseline ^a	82.38	79.73	63.52	79.76	72.44
		+ GIF	82.57 (+0.19)	86.06 (+6.33)	48.32 (-15.20)	85.62 (+5.86)	78.20 (+5.76)
		+ SE	82.78 (+0.40)	87.35 (+7.62)	47.06 (-16.46)	86.87 (+7.11)	79.21 (+6.77)
tiered-ImageNet	1-shot	Baseline ^a	71.19	75.96	70.13	76.25	69.14
		+ GIF	75.39 (+4.20)	79.95 (+3.99)	58.44 (-11.69)	79.70 (+3.45)	72.72 (+3.58)
		+ SE	77.26 (+6.07)	81.86 (+5.90)	55.29 (-14.84)	81.34 (+5.09)	74.31 (+5.17)
	5-shot	Baseline ^a	83.76	82.36	58.75	83.34	75.32
		+ GIF	84.89 (+1.13)	87.73 (+5.37)	41.16 (-17.59)	86.97 (+3.63)	80.19 (+4.87)
		+ SE	85.29 (+1.53)	88.67 (+6.31)	39.91 (-18.84)	87.86 (+4.52)	80.59 (+5.27)

^a Denotes ProtoNet with the proposed open loss.

Table 5

Comparison between semantic features and dimension-expanded visual features. Random denotes a fixed random projection of visual features to the same dimensionality, while VisualOnly denotes a learnable visual projection without semantic supervision.

Dataset	Method	1-shot		5-shot	
		ACC	AUROC	ACC	AUROC
mini-ImageNet	Random	71.66 $\pm$ 0.75	75.89 $\pm$ 0.66	80.92 $\pm$ 0.52	84.06 $\pm$ 0.43
	VisualOnly	72.89 $\pm$ 0.78	76.41 $\pm$ 0.70	81.98 $\pm$ 0.50	85.76 $\pm$ 0.40
	Ours	74.59$\pm$0.77	78.02$\pm$0.70	82.78$\pm$0.50	87.35$\pm$0.40
tiered-ImageNet	Random	74.75 $\pm$ 0.83	79.82 $\pm$ 0.64	83.85 $\pm$ 0.57	85.44 $\pm$ 0.48
	VisualOnly	76.29 $\pm$ 0.80	80.57 $\pm$ 0.64	84.45 $\pm$ 0.57	86.98 $\pm$ 0.49
	Ours	77.26$\pm$0.78	81.86$\pm$0.69	85.29$\pm$0.54	88.67$\pm$0.45
CUB-200-2011	Random	86.26 $\pm$ 0.66	90.13 $\pm$ 0.51	91.50 $\pm$ 0.32	94.21 $\pm$ 0.22
	VisualOnly	88.13 $\pm$ 0.61	91.17 $\pm$ 0.46	92.41 $\pm$ 0.33	95.32 $\pm$ 0.22
	Ours	89.35$\pm$0.59	92.10$\pm$0.45	93.47$\pm$0.29	96.45$\pm$0.20

Table 6

Comparison of different graph fusion strategies on mini-ImageNet under the 5-way 1-shot and 5-shot settings. V_f and S_f denote visual and semantic features, respectively.

Setting	Fusion Strategy	ACC $\uparrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUPRC $\uparrow$	F1 Score $\uparrow$
1-shot	V_f	72.99	76.34	64.45	77.13	69.77
	$[V_f \parallel S_f]$	73.20	77.14	63.98	77.94	70.66
	Ours	73.58	77.62	62.45	77.84	70.53
5-shot	V_f	82.07	85.99	48.92	85.52	78.07
	$[V_f \parallel S_f]$	81.62	85.44	49.72	85.06	77.32
	Ours	82.34	86.95	47.16	86.67	79.00

the performance gains are not merely due to increased dimensionality or additional trainable parameters. Instead, the semantic representations learned from image-description supervision offer complementary guidance to visual features, leading to improved performance in both closed-set recognition and open-set discrimination.

5.4.3. Analysis of graph fusion

Table 6 presents the results of different graph fusion strategies on mini-ImageNet. When using only visual features V_f , the model achieves a strong baseline performance, indicating that visual cues alone are effective for intra-class aggregation. However, directly concatenating visual and semantic features $[V_f \parallel S_f]$ brings limited improvements or even slight performance drops in some cases (e.g., lower AUROC and F1 Score in the 5-shot setting), likely due to feature misalignment or redundancy. In contrast, our method fuses V_f and S_f through element-

Table 7

Few-shot open-set recognition results (%) under different image description generation methods on mini-ImageNet.

Methods	mini-ImageNet			
	1-shot		5-shot	
	ACC	AUROC	ACC	AUROC
SP-CLIP (Chen et al., 2023)	72.89	76.73	82.69	86.39
SemFew (Zhang et al., 2024)	72.81	76.24	82.23	86.26
Ours	73.58	77.62	82.34	86.95

wise addition in a shared embedding space, which helps better alignment between modalities and more effective integration of semantic cues. Consequently, our method achieves the best results on most evaluation metrics under both the 1-shot and 5-shot settings, demonstrating the benefit of semantic-aware fusion in constructing more informative graph representations.

Although the improvement in ACC over direct feature concatenation is relatively modest, the proposed graph fusion strategy consistently improves AUROC, FPR95, and F1 Score. These metrics directly reflect the ability to distinguish known and unknown categories and are therefore particularly important in FSOR. Compared with direct concatenation, the proposed fusion strategy aligns visual and semantic representations in a shared embedding space before graph construction, resulting in more informative graph relationships and improved category separability. Furthermore, the graph is constructed at the episode level using a lightweight top- k nearest-neighbor strategy, resulting in limited additional computational overhead. Therefore, the proposed graph fusion strategy achieves a favorable balance between recognition performance and computational complexity.

5.4.4. Analysis of image descriptions

In Table 7, SP-CLIP achieves the highest 5-shot ACC, suggesting that its class-level textual prompts provide effective semantic guidance for closed-set classification when more support samples are available. In contrast, the proposed method achieves the best AUROC in both the 1-shot and 5-shot settings, as well as the highest ACC under the more challenging 1-shot scenario. This improvement can be attributed to the use of image-grounded semantic representations and multimodal graph fusion, which jointly capture semantic and structural relationships among samples and enhance the separation between known and unknown classes. Although SEM employs richer textual descriptions, the larger discrepancy between textual semantics and visual features make cross-modal alignment more difficult under limited samples, thereby limiting the effectiveness of feature fusion and open-set discrimination.

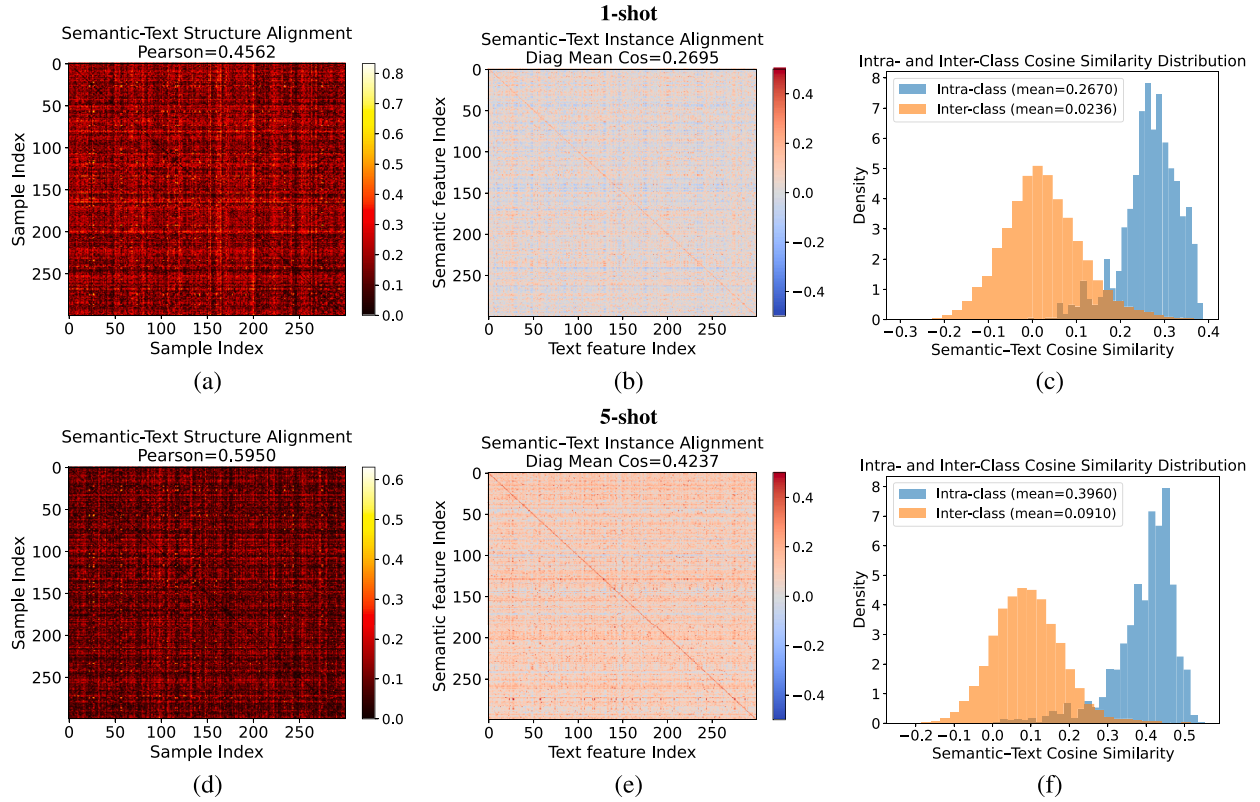


Fig. 4. Semantic-text feature alignment comparison under 1-shot and 5-shot settings on the test set of CUB-200-2011: (a,d) structural alignment heatmaps; (b,e) instance-level semantic-text similarity matrices; (c,f) intra- and inter-class semantic-text cosine similarity distributions.

To better understand how the number of support samples affects cross-modal representation learning, we further analyze the alignment between image-derived semantic features and text features on the CUB-200-2011 test set from structural, instance-level, and category-level perspectives. Structural alignment is evaluated by computing pairwise similarity matrices for semantic and text features, respectively, and measuring their consistency using Pearson correlation, which quantifies the linear correspondence between the two modalities. As illustrated in Fig. 4(a) and (d), the correlation increases from 0.4562 (1-shot) to 0.5950 (5-shot), indicating that the relational structure among samples becomes more consistent across modalities when more support samples are available. For instance-level alignment, we evaluate the diagonal mean cosine similarity between each image-derived semantic feature and its corresponding text feature, as presented in Fig. 4(b) and (e). The similarity increases from 0.2695 to 0.4237, demonstrating more accurate cross-modal matching under the 5-shot setting. For category-level alignment, we analyze the distributions of intra-class and inter-class cosine similarities between semantic and text features, as shown in Fig. 4(c) and (f). The intra-class similarity increases from 0.2670 (1-shot) to 0.3960 (5-shot), while inter-class similarity increases from 0.0236 to 0.0910. This is because additional support samples provide broader semantic coverage, leading to a denser embedding space. Nevertheless, the separation margin improves from 0.2434 to 0.3050, indicating enhanced category-level discriminability.

5.4.5. Robustness to description noise

To assess the impact of description quality, we further conduct a semantic corruption analysis on CUB-200-2011 by randomly replacing semantic attributes in the generated descriptions with incorrect alternatives while preserving the original sentence structure. For example, the description “black and grey plumage with a long beak” may be progressively corrupted as “black and grey fur with a long beak” (20%), “black and green fur with a long beak” (40%), and “white and green fur with a

Table 8

Robustness analysis under different semantic corruption ratios on CUB-200-2011.

Corruption Ratio	1-shot		5-shot	
	ACC	AUROC	ACC	AUROC
0%	89.35	92.10	93.47	96.45
20%	88.94	91.69	93.21	96.15
40%	88.58	91.34	92.90	95.82
60%	88.31	91.01	92.71	95.57
80%	88.08	90.66	92.52	95.21
100%	87.78	90.24	92.22	94.87

short beak” (60%). Table 8 shows that performance decreases progressively as the corruption ratio increases. Specifically, AUROC drops from 92.10% to 90.24% in the 1-shot setting and from 96.45% to 94.87% in the 5-shot setting when the corruption ratio reaches 100%. Despite the severe corruption, the performance degradation remains limited, suggesting that MGF benefits from fine-grained descriptions but does not overly rely on the correctness of individual semantic attributes. This result demonstrates the robustness of the proposed framework to semantic hallucinations and terminology errors in automatically generated descriptions.

5.4.6. Analysis of prompt templates

To investigate the sensitivity of MGF to prompt design, we compare the default prompt template adopted in our framework with four additional templates for generating image descriptions, covering declarative, attribute-oriented, and question-style formulations. The results are reported in Table 9. Overall, the performance variation across different prompts is relatively small, with maximum differences of only 0.80% in ACC and 0.86% in AUROC. This indicates that the proposed MGF

Table 9
Effect of different prompt templates on mini-ImageNet.

Prompt Template	1-shot		5-shot	
	Acc	AUROC	Acc	AUROC
A photo of a {class}. It has {description}.	74.16	77.84	82.62	86.97
Describe the visual attributes of this image.	74.21	77.81	82.39	86.54
Describe the color, shape and texture of the image.	74.96	77.82	82.93	86.62
Question: What are the visual characteristics of the image?	74.88	77.56	82.36	86.49
The {class} has a {description}. (ours)	74.59	78.02	82.78	87.35

Table 10
Few-shot open-set recognition results (%) under different adjacency matrix construction strategies on mini-ImageNet.

Strategies	mini-ImageNet			
	1-shot		5-shot	
	ACC	AUROC	ACC	AUROC
MLP (Satorras & Estrach, 2018)	73.31	77.76	81.05	86.29
Gaussian kernel (Liu et al., 2019)	72.11	75.84	80.68	84.11
Ours	73.58	77.62	82.34	86.95

framework is robust to prompt diversity and does not rely heavily on a specific prompt formulation.

Among the evaluated templates, the prompt “Describe the color, shape and texture of the image” achieves the highest classification accuracy under both the 1-shot and 5-shot settings, suggesting that emphasizing fine-grained visual attributes can provide more discriminative semantic cues. In contrast, the template “The {class} has a {description}” yields the best AUROC scores, indicating stronger capability in distinguishing known and unknown classes. These results also demonstrate that variations in the generated descriptions have limited impact on training performance, further showing the robustness of our framework to prompt variations. Since open-set recognition particularly emphasizes reliable unknown-class detection, we adopt this template in the final model.

5.4.7. Analysis of adjacency matrix

We consider two types of similarity-based adjacency constructions: MLP-based strategy (Satorras & Estrach, 2018) and Gaussian kernel-based strategy (Liu et al., 2019). The former employs a multilayer perceptron to compute pairwise similarities from the element-wise absolute differences between feature vectors, while the latter applies a Gaussian kernel with a learnable scaling factor to normalize feature distances. The comparison result is shown in Table 10. The MLP-based strategy achieves relatively competitive performance, particularly in the 1-shot setting, which can be attributed to the flexibility of the learnable adjacency matrix that captures absolute difference relationships between samples. Conversely, the Gaussian kernel-based strategy relies heavily on the learnable scaling factor; small estimation errors in this factor can lead to large variations in the adjacency matrix. This instability becomes more pronounced in novel FSOR tasks with limited samples, resulting in lower recognition performance. In contrast, our method explicitly computes similarities over concatenated visual and semantic features, followed by top- k nearest neighbor selection to determine meaningful connections. This avoids introducing additional learnable components, thereby reducing sensitivity to parameter tuning, while still achieving strong and consistent performance across different settings.

5.4.8. Complexity analysis

To assess the computational efficiency of the proposed framework, we compare the number of trainable parameters, GPU memory consumption, training time per episode, and inference time per episode with two representative FSOR methods, OPP and GEL, as well as several

ablated variants of our model. Since image descriptions are generated offline and reused across episodes, Table 11 excludes the caption generation cost and reports only the episodic training and inference cost of the recognition model. As shown in Table 11, GEL exhibits the highest computational cost, requiring substantially longer training and inference time together with significantly higher GPU memory consumption. This is mainly because GEL employs dual prediction branches and an energy-based optimization objective for open-set recognition. OPP also incurs additional computational overhead due to its prototype refinement strategy, which increases both parameter complexity and inference cost. In contrast, the proposed framework combines semantic extraction and graph-based feature fusion within a relatively lightweight architecture and does not rely on auxiliary optimization procedures or additional prediction branches. Notably, compared with the baseline model, the full MGF framework increases the training time by only 10 ms per episode and the peak GPU memory by only 0.10 GB, indicating that the proposed modules introduce only marginal overhead while maintaining strong recognition performance.

The ablation results further illustrate the impact of each component on efficiency. Introducing the semantic extraction (SE) module increases the number of trainable parameters due to the image-to-semantic alignment network, leading to a modest increase in memory usage and running time. In contrast, the graph information fusion (GIF) module adds only a limited number of parameters but requires additional graph construction and message propagation operations, resulting in higher training and inference time. When both components are integrated, the full model naturally exhibits the highest computational cost among the proposed variants. Nevertheless, the increase remains well controlled and is still lower than or comparable to that of existing FSOR approaches such as OPP and GEL. Overall, the proposed framework achieves consistent performance improvements while introducing only limited computational overhead in terms of model size, memory usage, and execution time.

5.4.9. Analysis of number of neighbors

Fig. 5 illustrates the effect of the number of nearest neighbors k on both closed- and open-set performance under the 5-shot setting. As k increases from 1 to 5, both metrics gradually improve, with AUROC increasing from 80.92% to 86.95% and ACC improving from 77.81% to 82.34%. This suggests that using an appropriate number of neighbors allows the model to capture more relevant samples and enhance the discrimination of each sample. When k exceeds 5, the AUROC performance begins to decline and decreases slightly to 85.48% at $k = 15$. This suggests that an excessively large number of neighbors tends to introduce more semantically similar samples from different classes, which hinders the model from obtaining more discriminative class representations in open-set scenarios from known and unknown queries. Therefore, we select $k = 5$ to construct the graph in all experiments.

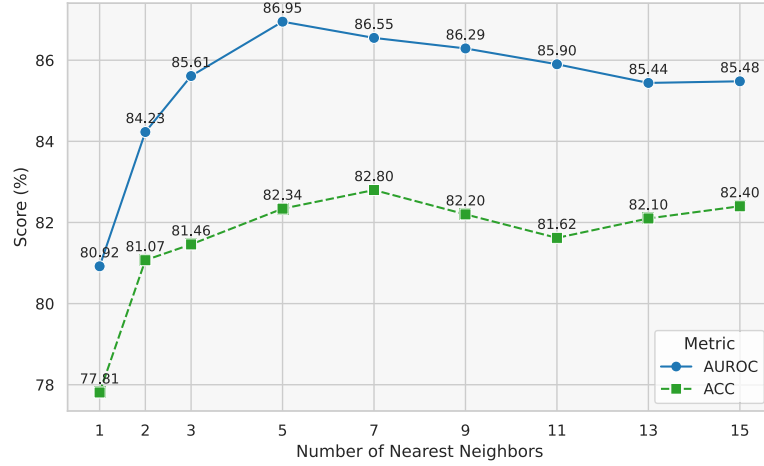
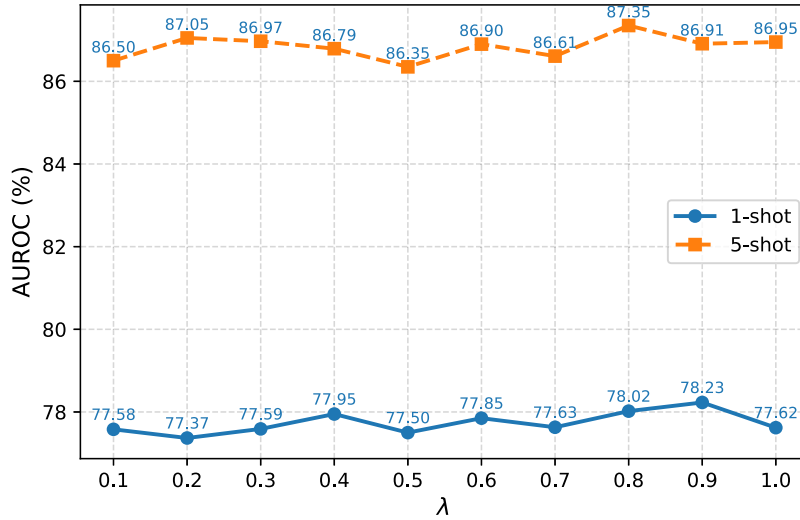
5.4.10. Analysis of loss function

The scaling factor λ controls the magnitudes of the contrastive loss, which together with the closed-set classification and alignment losses influences the optimization process. To determine the optimal scaling factor, we conducted experiments on mini-ImageNet for 1- and 5-shot

Table 11

Comparison of model complexity, computational efficiency, and memory consumption.

Group	Method	Parameters (M)	Training Time (ms/episode)	Inference Time (ms/episode)	Peak GPU Memory (GB)
Reference Methods	OPP	18.08	245	80	7.90
	GEL	14.06	490	133	20.03
MGF Variants	Baseline	12.42	218	41	7.04
	+ GIF	13.98	223	43	7.07
	+ SE	16.91	225	44	7.12
	MGF	18.47	228	44	7.14

**Fig. 5.** Performance comparison under different numbers of nearest neighbors on 5-shot tasks.**Fig. 6.** AUROC performance of 1-shot and 5-shot settings under different values of scaling factor λ .

tasks, and the results are shown in Fig. 6. For the 1-shot task, the AUROC varies slightly within a narrow range, from 77.37% to 78.23%, indicating that the model performance is relatively insensitive to λ under extremely low-shot conditions. The best AUROC (78.23%) is achieved at $\lambda = 0.9$, but the overall variation is less than 1%. For the 5-shot task, the AUROC consistently remains above 86%, ranging from 86.32% to 87.35%, suggesting that the performance is slightly more sensitive to λ compared to the 1-shot case. The highest AUROC is observed at $\lambda = 0.8$. Therefore, this value was chosen as the optimal scaling factor.

5.4.11. Visualization of score distributions

Fig. 7 compares the normalized open-set score distributions for known and unknown classes using four representative methods on mini-ImageNet. An ideal open-set recognition model should assign low scores to unknown samples and high scores to known ones, resulting in clearly separated distributions. Among the methods, our approach shows the lowest IoU of 0.29, indicating the smallest overlap between known and unknown score distributions. This suggests that our model effectively distinguishes between known and unknown

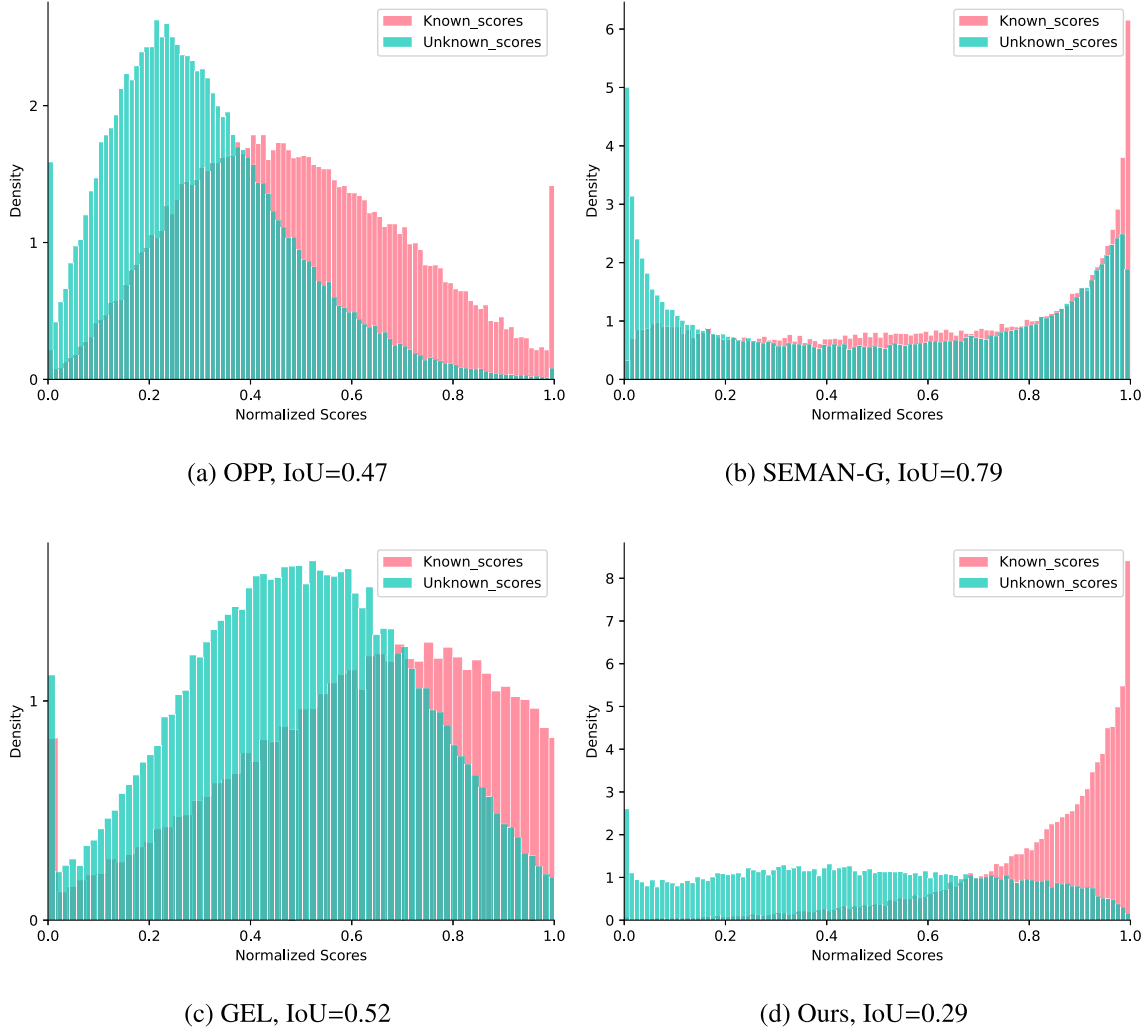


Fig. 7. Visualization of normalized open-set score distribution for known and unknown samples on mini-ImageNet for 5-way 5-shot tasks. IoU indicates the ratio of the intersection of two distributions.

classes through multimodal information fusion. In contrast, SEMAN-G fuses the visual features with textual embeddings through attention, but has the highest IoU of 0.79, indicating the model fails to handle the semantic gaps between the two modalities, leading to substantial overlap and poor separation. GEL adopts the meta-trained model of FEAT (Ye et al., 2020) to improve open-set recognition; however, it still suffers from difficulty in distinguishing known from unknown classes with a large IoU of 0.52. OPP has a slightly smaller IoU (0.47) compared to SEMAN-G and GEL owing to the learning of global class embeddings.

6. Conclusion

In this study, we proposed a multimodal graph-based fusion (MGF) framework to address few-shot open-set recognition (FSOR) by learning visually grounded semantic representations under the supervision of image descriptions. Unlike prior works that require textual input at inference, MGF learns semantically aligned representations from visual features under textual supervision during training, enabling semantic-aware inference without requiring textual inputs during testing. To improve discrimination between known and unknown classes, we employ a graph convolutional network to perform multi-modal feature fusion between the visual and semantic features for effective intra-class information propagation. Extensive experiments on

standard few-shot benchmarks validate the effectiveness of each component and demonstrate the superiority of MGF over existing methods for FSOR.

Despite its strong performance, the proposed MGF framework has certain limitations. The quality of the semantic supervision is inherently tied to the image descriptions generated by the vision-language model (e.g., BLIP-2). Although we employ an induced prompt to mitigate this, the descriptions may still contain inaccuracies or lack fine-grained detail for highly specialized or uncommon visual concepts, which can introduce noise into the semantic targets. Furthermore, the graph-based fusion module, while effective, relies on the construction of a task-specific graph based on a fixed number of nearest neighbors. This predefined structure may not be optimal for all tasks, especially when the intra-class variance is high or the inter-class similarity is significant, potentially leading to suboptimal message propagation. In future work, we aim to address these limitations from several directions. First, we plan to explore more robust methods for generating semantic supervision, such as leveraging multi-modal large language models to produce hierarchical or part-based descriptions, or incorporating a noise-filtering mechanism to select only the most reliable generated captions. Second, we intend to investigate adaptive graph construction strategies, where the graph structure (e.g., the number of neighbors or edge weights) can be learned dynamically based on the characteristics of each task. Finally, we are interested in extending MGF to more challenging cross-domain few-shot

open-set scenarios, where the distribution shift between base and novel classes is more severe, to further test the generalization capability of our framework.

CRediT authorship contribution statement

Xilang Huang: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization; **Seon Han Choi:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets supporting this study are publicly available and can be accessed from their official sources as cited in the paper.

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